

STANDARD OPERATING PROCEDURE

Canvas Discussion Grader

Grading Distribution Tool — User Guide

Document Version	1.0
Effective Date	Spring 2026
Prepared By	University of Pittsburgh — TA Staff
Application	https://web-production-a3be2.up.railway.app/

1. Purpose & Overview

This SOP describes the procedure for using the Canvas Discussion Grader to analyze student participation in Canvas discussions and produce grade distributions based on a normal distribution model.

The tool connects to the Canvas LMS via API, retrieves all discussion post data, and outputs a graded breakdown per student — downloadable as Excel, CSV, PNG, and log files.

2. Generating a Canvas API Token

2.1 Access Canvas Settings

1. Log in to Canvas at your institution's URL.
2. Click **Account** in the left navigation, then select **Settings**.

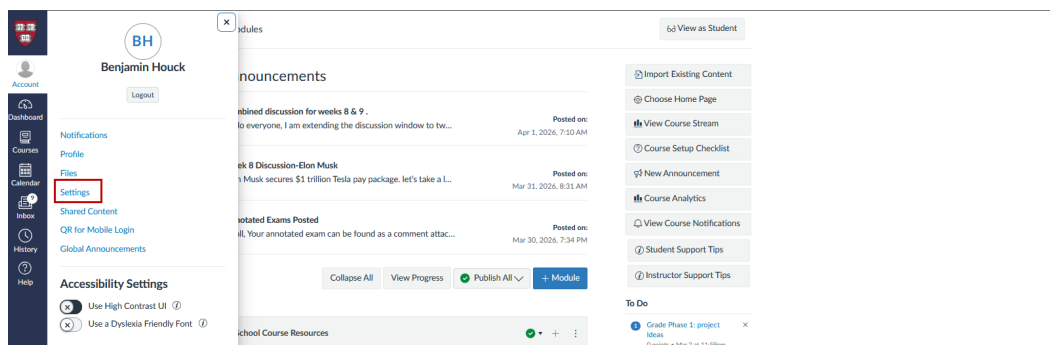


Figure 1 — Canvas account menu (Account → Settings highlighted)

2.2 Create a New Access Token

3. Scroll down to the **Approved Integrations** section.
4. Click **+ New Access Token**.

The screenshot shows the Canvas user interface. On the left is a sidebar with navigation links: Account, Dashboard, Courses, Calendar, Inbox, History, and Help. The main content area is titled 'Approved Integrations' and contains a table of authorized third-party applications. The table has columns for App, Status, Purpose, Dates, Details, and Remove. The rows include 'Hello Harvard', 'SensusAccess Convert a File', and 'User-Generated'. Below the table, the '+ New Access Token' button is highlighted with a red box. The 'Feature Options' section below the button includes a search bar and a 'Clear' button.

App	Status	Purpose	Dates	Details	Remove
Hello Harvard	active	Hello Harvard	Expires: never Last Used: Jan 19 at 11:36am	details	
SensusAccess Convert a File	active	SensusAccess Convert a File	Expires: never Last Used: Feb 7 at 2:27pm	details	
User-Generated	active	Grading optimization	Expires: never Last Used: Apr 9 at 1:10am	details	

Figure 2 — Approved Integrations section with New Access Token highlighted

5. In the **Purpose** field, enter a descriptive label such as "Grading Optimization".
6. Leave the date fields blank to create a non-expiring token.
7. Click **Generate Token**.

The screenshot shows the 'New Access Token' dialog box. It contains a text field for 'Purpose', two date/time fields for 'Expiration date' and 'Expiration time', and a 'Generate Token' button highlighted with a red box. The dialog also includes a 'Cancel' button and a brief explanation of access tokens.

Figure 3 — Generate Token button highlighted

Note: Copy and save the token immediately — it will not be shown again after closing this dialog.

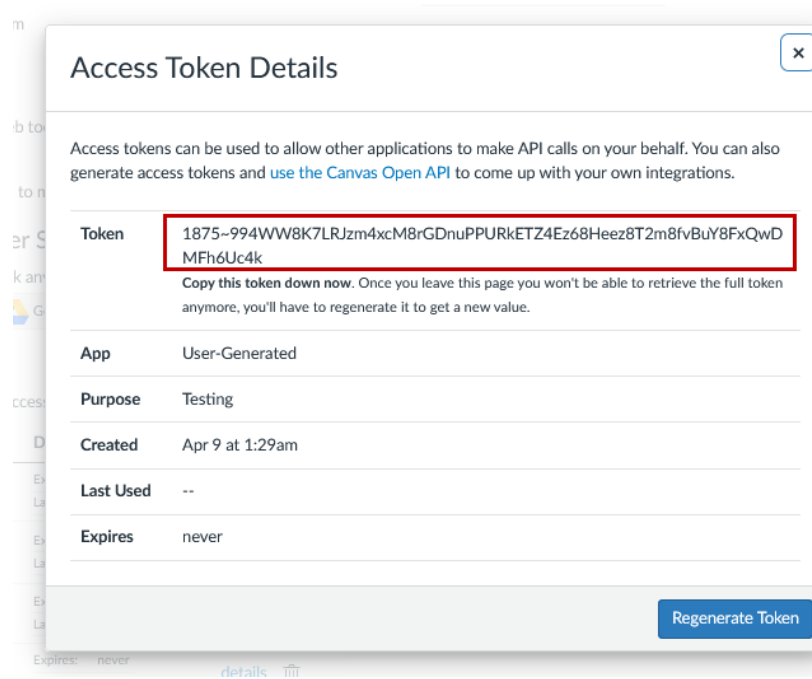


Figure 4 — Token value field to copy and save

3. Connecting to the Grader Tool

3.1 Access the Application

1. Navigate to the application URL:

<https://web-production-a3be2.up.railway.app/>

2. Enter the following in the sidebar connection panel:

- Canvas URL (e.g., <https://pitt.instructure.com>)
- API Access Token (generated in Section 2)
- Course ID (see Section 3.2 below)

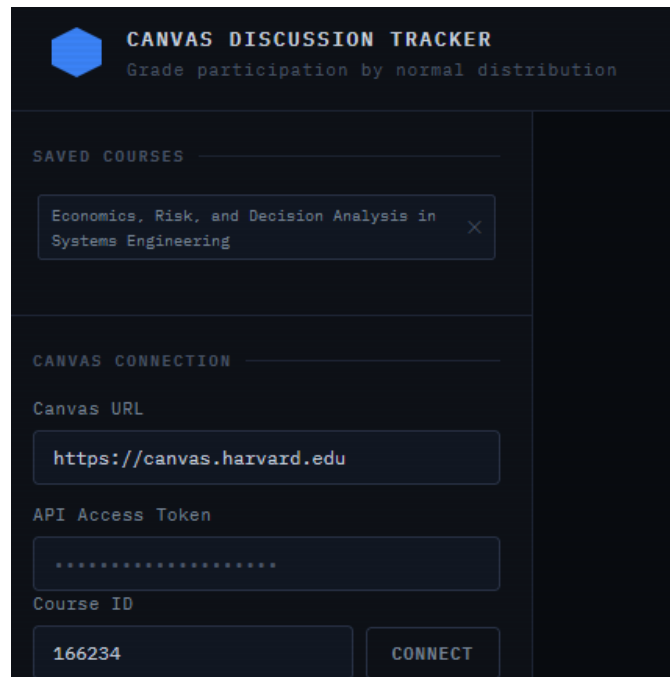
The screenshot shows a dark-themed sidebar panel titled "CANVAS DISCUSSION TRACKER" with the subtitle "Grade participation by normal distribution". It contains a "SAVED COURSES" section with a single entry: "Economics, Risk, and Decision Analysis in Systems Engineering". Below this is the "CANVAS CONNECTION" section, which includes input fields for "Canvas URL" (containing "https://canvas.harvard.edu"), "API Access Token" (masked with dots), and "Course ID" (containing "166234"). A "CONNECT" button is located at the bottom right of the connection section.

Figure 5 — Connection panel in sidebar

3.2 Locate Your Course ID

The Course ID is the number in the Canvas course URL. For example, in <https://pitt.instructure.com/courses/123456>, the Course ID is **123456**.

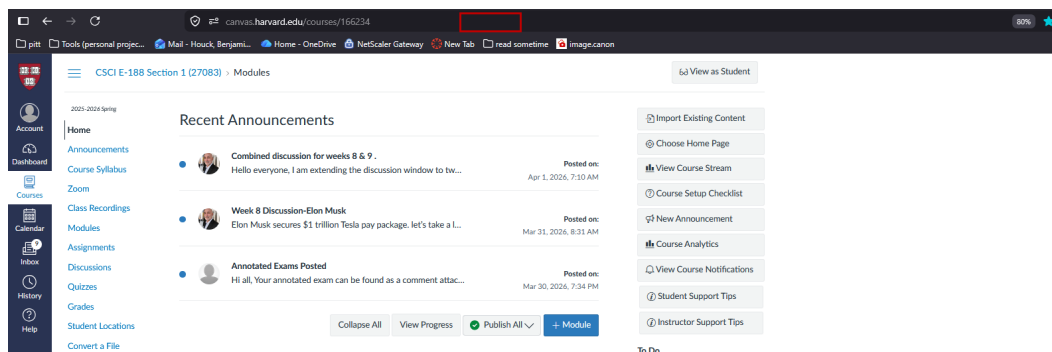
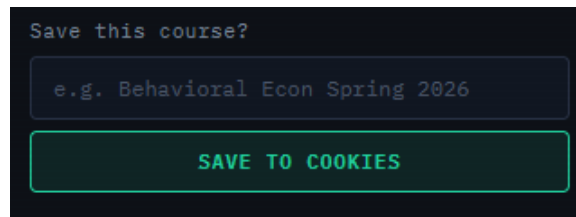


Figure 6 — Course ID location in the Canvas URL bar highlighted

3.3 Connect and Save

3. Click **Connect**.
4. Once connected, enter a course alias (e.g., "*Behavioral Econ Spring 2026*") and click **Save to Cookies** for quick future access.

A dark-themed modal window titled "Save this course?". It contains a text input field with the placeholder text "e.g. Behavioral Econ Spring 2026" and a prominent green button labeled "SAVE TO COOKIES".

Save this course?

e.g. Behavioral Econ Spring 2026

SAVE TO COOKIES

Figure 7 — Save to Cookies panel

4. Configuring the Analysis

4.1 Omitting Discussions

By default the tool includes all discussion topics in the course. Use the **Omit Discussions** panel to exclude specific topics before running.

1. Use the search bar to filter topics by name.
2. Click individual topics to toggle omission.
3. Click **Omit visible** to exclude all topics currently shown in the search results (e.g., to exclude all Presentation topics at once).

The screenshot shows the 'GRADING SCHEME' panel with a table of conditions and grades. The table has two columns: 'CONDITION (Z-SCORE)' and 'GRADE %'. There are five rows of conditions, each with a colored dot, a condition description, a z-score input, a grade percentage input, and a delete button (X).

CONDITION (Z-SCORE)	GRADE %
z ≥ 1	120
z ≥ 0	100
z ≥ -1	100
z ≥ -2	80
else (below all)	0

Below the table is a '+ Add tier' button. At the bottom, there is a summary list of the conditions and their corresponding grades:

- z ≥ 1: 120%
- z ≥ 0: 100%
- z ≥ -1: 100%
- z ≥ -2: 80%
- otherwise: 0%

Figure 8 — Omit Discussions panel

4.2 Excluding Students

Students can be excluded from the calculation (e.g., students who have dropped).

4. Use the search bar under **Exclude Students**.
5. Click individual students to toggle, or use **Exclude visible** / **Clear visible** for bulk actions.
6. Hold **Shift** and click to select a contiguous range.

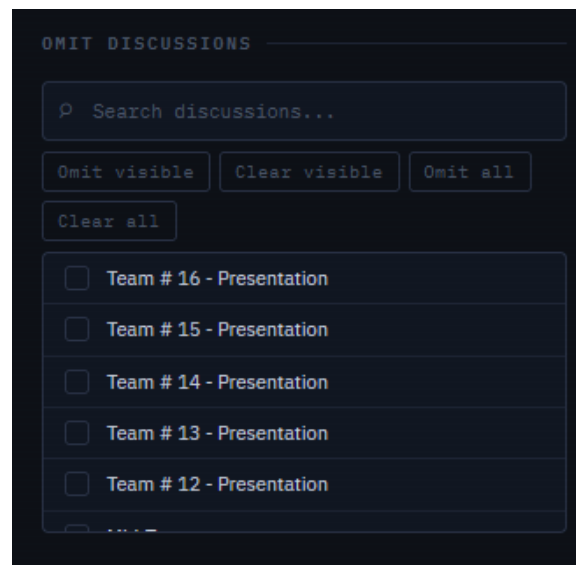


Figure 9 — Exclude Students panel

4.3 Grading Scheme

The **Grading Scheme** section lets you configure custom z-score thresholds for each grade tier.

- Default: $z \geq 1 \rightarrow 120\%$, $z \geq 0 \rightarrow 100\%$, $z \geq -1 \rightarrow 80\%$, else $\rightarrow 0\%$
- Tiers can be added, removed, or edited before each run.
- Click **+ Add tier** to insert a new threshold row.

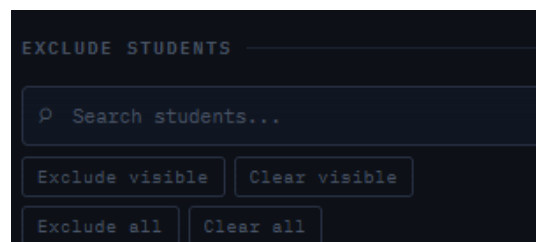


Figure 10 — Grading Scheme panel

Note: Apply the grading schema that reflects the course syllabus. Spring 2026 used the default four-tier schema above.

5. Running the Analysis

1. Verify all filters and the grading scheme are configured correctly.
2. Click **Run Analysis** in the sidebar.

A progress bar will display while the tool fetches and processes discussion data. Upon completion, the main panel updates with the full results.

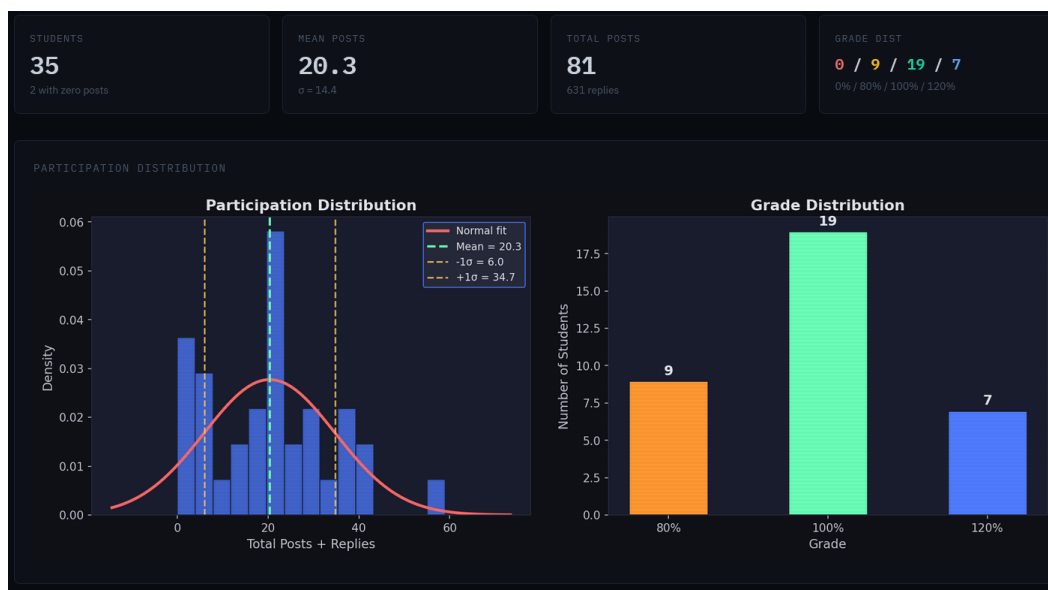


Figure 11 — Results panel after a completed run

6. Interpreting the Output

6.1 Summary Cards

Four summary cards appear at the top of the results panel:

- **Students** — total count and number with zero posts
- **Mean Posts** — average participation $\pm$ standard deviation
- **Total Posts** — aggregate post and reply counts
- **Grade Distribution** — count per grade tier

6.2 Distribution Chart

A histogram overlaid with the fitted normal curve is displayed. Dashed vertical lines mark the mean (green) and the $\pm 1\sigma$ boundaries (yellow), corresponding to grade tier cutoffs.

7. Downloading Results

All output files are timestamped (YYYYMMDD_HHMMSS) to prevent accidental overwriting. Files can be downloaded individually or as a single **.zip** archive via the Export Files section.

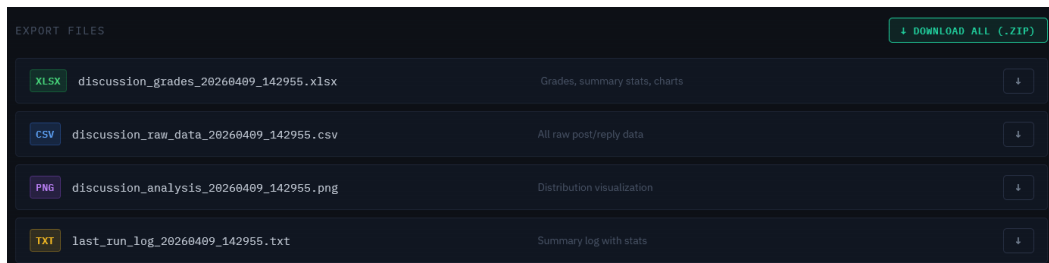


Figure 12 — Export Files section showing timestamped downloads

Type	Filename	Contents
XLSX	discussion_grades_{ts}.xlsx	Grades, z-scores, summary stats, embedded charts
CSV	discussion_raw_data_{ts}.csv	Raw post/reply data for auditability
PNG	discussion_analysis_{ts}.png	Participation distribution histogram image
TXT	last_run_log_{ts}.txt	Mean, std dev, grade breakdown, zero-post list

7.1 Student Results Table

Below the download section, a scrollable table shows every student with their post count, reply count, total, z-score, and assigned grade tier.

STUDENT RESULTS						
						35 students
#	STUDENT	POSTS	REPLIES	TOTAL	Z-SCORE	GRADE
1	Denisa Tutunaru-Stan	5	54	59	2.687	120%
2	Angel Albornoz	3	37	40	1.366	120%
3	Augustin Gaset	0	40	40	1.366	120%
4	Oscar Herrera Beltran	4	35	39	1.297	120%
5	Lindsey Saunders	0	39	39	1.297	120%
6	Allie DeGuzman	6	30	36	1.088	120%
7	Shaun Rodman	5	30	35	1.019	120%

Figure 13 — Per-student results table